





Proactive Routing

1. INITIALIZATION PHASE:
 - NODES EXCHANGE INFORMATION ABOUT THEIR NEIGHBORS AND ESTABLISH A ROUTING TABLE.
 - EACH NODE MAINTAINS A TOPOLOGY MAP OF THE NETWORK.
2. PERIODIC UPDATES:
 - NODES PERIODICALLY EXCHANGE CONTROL MESSAGES TO UPDATE ROUTING TABLES.
 - THESE UPDATES ENSURE THAT ROUTING INFORMATION REMAINS CURRENT DESPITE CHANGES IN NETWORK TOPOLOGY OR NODE MOBILITY.
3. EFFICIENCY:
 - PROACTIVE PROTOCOLS ARE EFFICIENT FOR NETWORKS WITH RELATIVELY STABLE TOPOLOGIES.
 - THEY PROVIDE QUICK DATA TRANSMISSION AS ROUTES ARE ALREADY ESTABLISHED.



Reactive Routing

1. ROUTE DISCOVERY:
 - WHEN A NODE NEEDS TO COMMUNICATE WITH ANOTHER NODE, IT INITIATES A ROUTE DISCOVERY PROCESS.
 - THE SOURCE NODE BROADCASTS A ROUTE REQUEST (RREQ) PACKET TO ITS NEIGHBORS.
2. ROUTE ESTABLISHMENT:
 - UPON RECEIVING THE RREQ, INTERMEDIATE NODES RECORD THE ROUTE BACK TO THE SOURCE.
 - THE DESTINATION NODE OR AN INTERMEDIATE NODE WITH A ROUTE TO THE DESTINATION SENDS A ROUTE REPLY (RREP) BACK TO THE SOURCE.
3. EFFICIENCY:
 - REACTIVE PROTOCOLS ARE SUITABLE FOR DYNAMIC NETWORKS WHERE NODES FREQUENTLY CHANGE POSITIONS.
 - THEY CONSERVE NETWORK RESOURCES BY ESTABLISHING ROUTES ONLY WHEN NEEDED.

